

Rishi Patel

Software Engineer | MERN Stack Developer

Contact

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About Me

Computer Science undergraduate with hands-on experience building full-stack web applications using React.js, Node.js, Express.js, and MongoDB. Developed SaaS and automation solutions focused on solving real-world problems. Focused on backend development, scalable systems, and software engineering.

Skills

Programming Languages: Java, JavaScript, Python

Frontend: HTML5, CSS3, React.js, Tailwind CSS

Backend: Node.js, Express.js, REST APIs

Database: MongoDB

Tools: Git, GitHub

Cloud & Deployment: Render, Vercel, Netlify, MongoDB Atlas

Computer Science Fundamentals: OOP, DBMS, Software Engineering

Key Achievements

- Secured 2nd Position at SPARK-2026 National-Level Project Competition.
- Reached semifinals in the 2025 National Build with India Hackathon

Certifications

- Postman API Fundamentals Student Expert
- Full Stack Developer Bootcamp - GeeksforGeeks

Experience

- Full Stack Web Developer** July 2025- Present
Freelance
 - Developed and deployed 3+ custom full-stack web applications using React.js, Node.js, Express.js, and MongoDB for freelance clients.
 - Designed responsive user interfaces, REST APIs, authentication, and database integrations, achieving 95% client satisfaction and consistent on-time delivery.
- Web Developer Intern** Feb 2025- March 2025
VirtuNexa
 - Contributed to 2+ scalable full-stack web applications using React.js, Node.js, Express.js, and MongoDB while following clean architecture principles.
 - Built 10+ REST APIs with JWT authentication and role-based access control (RBAC), enhancing application security and user authorization.

Projects

SmartTable - SaaS Timetable Management Platform

Tech Stack: React.js · Node.js · Express.js · MongoDB · JWT · Brevo

- Architected a multi-tenant SaaS platform with 4 user roles (Admin, Scheduler, Teacher, Student) and 100% institution-level data isolation.
- Created a Stripe-based subscription system with a 14-day trial, Standard/Flex plans, and automated renewals, payment failure management, and account lifecycle management.
- Engineered a real-time drag-and-drop timetable generation engine with conflict detection for teacher availability, improving scheduling accuracy by over 90%.
- Built 15+ REST APIs for authentication, timetable scheduling, user management, and subscription workflows using Express.js and MongoDB.
- Implemented JWT (JSON Web Token) authentication, email verification, password reset, resend verification, and Brevo Email API integration to strengthen platform security and onboarding.

Automated Floor Plan Generator for Irregular Polygonal Plots

Tech Stack: React.js · Node.js · FastAPI · Python · Shapely

- Engineered a geometry-processing engine for arbitrary polygonal inputs using Shapely, enabling automated spatial analysis and floor plan generation.
- Designed and implemented Recursive Binary Space Partitioning (BSP) algorithms for intelligent room allocation across irregular plot geometries.
- Built a constraint-based layout optimization workflow that validates room dimensions, adjacency requirements, circulation paths, and architectural feasibility during plan generation.
- Achieved 78-91% space utilization efficiency across rectangular, pentagonal, hexagonal, and trapezoidal plot geometries using geometry-aware optimization.
- Developed FastAPI-based REST APIs and integrated them with a React.js frontend through a Node.js gateway, enabling real-time floor plan generation and visualization.

Education

- Computer Science and Engineering** 2023-2027
K.D.K College of Engineering, Nagpur
CGPA: 7.22/10
- Higher Secondary Certificate (HSC)** 2022-2023
Bharti Krishna Vidya Vihar, Nagpur
Percentage: 65%
- Secondary School Certificate (SSC)** 2020-2021
Bharti Krishna Vidya Vihar, Nagpur
Percentage: 75%

View more projects: [Rishi's Portfolio](#)